

C907

Introduction to Biology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A community newsletter written by retired engineers states that cell theory is “only a theory, so it is basically a guess.” Which response best reflects how scientists use the word theory?

- A. A theory is a broad explanation supported by many tested hypotheses and observations.
- B. A theory and a hypothesis mean the same thing.
- C. It will become a law once it is proven.
- D. A scientific theory can never be revised.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which statement is a testable hypothesis?

- A. All cells come from preexisting cells.
- B. Plants are the most beautiful organisms.
- C. Bean plants given 8 hours of light a day will grow taller in 3 weeks than those given 4 hours.
- D. Plants grow because an unseen life force wants them to.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A microbe from a hot spring in a national park, where visitors number 4 million a year, has DNA in a nucleoid region, ribosomes, a cell wall, and no membrane-bound organelles. How is it classified?

- A. A virus
- B. Prokaryote
- C. Eukaryote, because it has ribosomes
- D. A plant cell, since any cell with a wall is a plant

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A muscle cell in a marathon runner who trains in a city 1,600 m above sea level contracts. What directly powers the motor proteins?

- A. Breaking ATP into ADP and phosphate
- B. Building ATP from ADP and phosphate
- C. Glucose, used directly by the proteins
- D. Oxygen, used directly by the proteins

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A nerve cell and a skin cell from the same person, both sampled at a clinic that just added evening hours, look and act very differently. Why?

- A. Only nerve cells carry the complete genome.
- B. They have the same DNA but express different genes.
- C. Nerve cells have lost the genes for skin proteins.
- D. Each cell type has its own separate genome.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A nucleus from a specialized frog intestine cell is placed into an egg whose own nucleus was removed, and a tadpole develops. What does this show?

- A. Intestine cells carry only intestine genes.
- B. Specialized cells still contain the full set of genes.
- C. The egg's cytoplasm created the missing genes.
- D. Any cell from any species can always grow a new organism.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A breeder at a nursery that ships to 12 states crosses red snapdragons (RR) with white (WW), and all offspring are pink. The breeder then crosses two pink plants. What ratio of offspring is expected?

- A. Red-and-white spotted flowers
- B. 3 red : 1 white
- C. All pink
- D. 1 red : 2 pink : 1 white

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A red bull and a white cow produce roan calves whose coats have both red hairs and white hairs. What pattern of inheritance is this?

- A.** Complete dominance of red
- B.** Blending, in which the alleles merge permanently
- C.** Incomplete dominance
- D.** Codominance

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A mother who carries hemophilia ($X^H X^h$) and an unaffected father ($X^H Y$), who run a bakery together, ask about risk. What fraction of their sons are expected to have hemophilia?

- A.** All of them
- B.** 1/4
- C.** None, because the father is unaffected
- D.** 1/2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

In a moth population on an island reached only by ferry, 16% show the recessive white wing color. Assuming Hardy-Weinberg equilibrium, what percent are heterozygous carriers?

- A.** 84%
- B.** 40%
- C.** 48%
- D.** 24%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

In a population in Hardy-Weinberg equilibrium, the dominant allele frequency is 0.7. What fraction shows the recessive phenotype?

- A.** 0.42
- B.** 0.3
- C.** 0.09
- D.** 0.49

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

The forelimbs of a whale, a bat, and a human, displayed in a museum hall that reopened last year, contain the same bones arranged differently and are used for swimming, flying, and grasping. What are they?

- A.** Homologous structures showing common ancestry
- B.** Evidence that the three evolved independently
- C.** Analogous structures
- D.** Vestigial structures

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A community newsletter ~~written by retired engineers~~ states that cell theory is “only a theory, so it is basically a guess.” Which response best reflects how scientists use the word theory?

Target word: meaning of scientific theory **Decides:** A theory is a well-tested explanation.

A	Correct	That is the scientific meaning.
B	Adjacent category	A hypothesis is a single testable prediction.
C	Confusion set	Laws describe patterns; theories explain them. One does not turn into the other.
D	Absolute language	Theories are refined when new evidence appears.

2. Answer C Concept 1 • transfer

Which statement is a testable hypothesis?

Target word: spot a testable prediction **Decides:** A measurable, falsifiable outcome.

A	Confusion set	A principle of a broad theory, not a single experimental prediction.
B	Adjacent category	A value judgment, not a test.
C	Correct	Specific, measurable, and could be proven wrong.
D	Dropped element	Drops anything measurable; it cannot be tested.

3. Answer B Concept 2

A microbe from a hot spring ~~in a national park, where visitors number 4 million a year,~~ has DNA in a nucleoid region, ribosomes, a cell wall, and no membrane-bound organelles. How is it classified?

Target word: cell category **Decides:** No nucleus and no membrane-bound organelles.

A	Adjacent category	Viruses lack ribosomes and are not cells.
B	Correct	A nucleoid instead of a nucleus marks a prokaryote.
C	Confusion set	Both cell types have ribosomes.
D	Absolute language	Bacteria, archaea, and fungi also have walls.

4. Answer A Concept 3

A muscle cell in a marathon runner ~~who trains in a city 1,600 m above sea level~~ contracts. What directly powers the motor proteins?

Target word: direct energy source **Decides:** ATP is the cell’s working energy currency.

A	Correct	ATP hydrolysis releases the energy motor proteins use.
B	Inverted operation	Making ATP stores energy; it does not release it.
C	Adjacent category	Glucose energy must first be captured in ATP.
D	Confusion set	Oxygen is the final electron acceptor in respiration, not a fuel.

5. Answer B Concept 4

A nerve cell and a skin cell from the same person, ~~both sampled at a clinic that just added evening hours,~~ look and act very differently. Why?

Target word: cause of specialization **Decides:** Same genome, different expression.

A	Absolute language	Nearly all body cells carry the full genome.
B	Correct	Gene regulation shapes each cell type.
C	Confusion set	Unused genes are silenced, not removed.
D	Adjacent category	All body cells come from one fertilized egg.

6. Answer B Concept 4 • transfer

A nucleus from a specialized frog intestine cell is placed into an egg whose own nucleus was removed, and a tadpole develops. What does this show?

Target word: what the experiment proves **Decides:** A specialized nucleus can direct a whole organism.

A	Inverted operation	The result shows the opposite.
B	Correct	All genes needed for a tadpole were present.
C	Adjacent category	Cytoplasm regulates genes; it does not create them.
D	Absolute language	Cloning is rare and often fails.

7. Answer D Concept 5

A breeder ~~at a nursery that ships to 12 states~~ crosses red snapdragons (RR) with white (WW), and all offspring are pink. The breeder then crosses two pink plants. What ratio of offspring is expected?

Target word: F2 phenotype ratio **Decides:** Pink is a blended heterozygote: incomplete dominance.

A	Adjacent category	Both colors showing fully is codominance.
B	Confusion set	That is complete dominance.
C	Dropped element	Drops the separation of alleles in the parents.
D	Correct	RW × RW gives RR, 2 RW, WW.

8. Answer D Concept 5 • transfer

A red bull and a white cow produce roan calves whose coats have both red hairs and white hairs. What pattern of inheritance is this?

Target word: name the pattern **Decides:** Both traits appear fully, side by side.

A	Adjacent category	White would then be hidden.
B	Absolute language	Alleles stay separate and can reappear.
C	Confusion set	That would give an intermediate color on each hair.
D	Correct	Both alleles are fully expressed.

9. Answer D Concept 6

A mother who carries hemophilia ($X^H X^h$) and an unaffected father ($X^H Y$), ~~who run a bakery together~~, ask about risk. What fraction of their sons are expected to have hemophilia?

Target word: risk among sons **Decides:** Sons get their X from the mother; half get X^h .

A	Absolute language	The mother has one normal X to pass on.
B	Confusion set	That is the chance for all children, not sons.
C	Dropped element	Drops the mother's X, the one sons inherit.
D	Correct	Each son has a 50% chance of inheriting X^h .

10. Answer C Concept 7

In a moth population on an island ~~reached only by ferry~~, 16% show the recessive white wing color. Assuming Hardy-Weinberg equilibrium, what percent are heterozygous carriers?

Target word: carrier frequency **Decides:** $q^2 = 0.16$, so $q = 0.4$ and $p = 0.6$.

A	Confusion set	That is everyone with the dominant phenotype.
B	Adjacent category	That is the allele frequency q .
C	Correct	$2pq = 2(0.6)(0.4) = 0.48$.
D	Dropped element	pq without the 2.

11. Answer C Concept 7 · transfer

In a population in Hardy-Weinberg equilibrium, the dominant allele frequency is 0.7. What fraction shows the recessive phenotype?

Target word: recessive phenotype from p **Decides:** $q = 0.3$; phenotype = q^2 .

A	Confusion set	$2pq$ is the carriers.
B	Dropped element	Forgets to square q .
C	Correct	0.3^2 .
D	Inverted operation	p^2 is homozygous dominant.

12. Answer A Concept 8

The forelimbs of a whale, a bat, and a human, ~~displayed in a museum hall that reopened last year~~, contain the same bones arranged differently and are used for swimming, flying, and grasping. What are they?

Target word: structure type and meaning **Decides:** Same bone plan, different functions.

A	Correct	Shared plan from a common ancestor.
B	Inverted operation	Shared structure points to shared ancestry.
C	Confusion set	Analogous structures share function, not origin.
D	Adjacent category	These limbs are fully functional.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Hypothesis vs. theory	A hypothesis is a testable, falsifiable prediction; a scientific theory is a broad, well-tested explanation that ties together many observations.	Treating “theory” as a guess, or thinking theories graduate into laws.	“Just a theory,” or “Which statement is testable?”
2	Prokaryotic vs. eukaryotic cells	Prokaryotes lack a nucleus and membrane-bound organelles; both cell types have ribosomes, a membrane, and DNA.	Using ribosomes or a cell wall to decide the category.	A list of cell parts and “classify the cell”.
3	ATP and energy coupling	Cells power work by hydrolyzing ATP to ADP + phosphate; food energy is first captured in ATP.	Thinking glucose or oxygen directly powers cell work, or that making ATP releases energy.	“What directly powers...?”
4	Differentiation through gene expression	Nearly all body cells carry the same genome; cells specialize by turning different genes on and off.	Believing specialized cells lose the genes they do not use.	Two cell types from one body that look and act differently.
5	Incomplete dominance vs. codominance	Incomplete dominance gives a blended intermediate (pink); codominance shows both traits fully (roan, AB blood). A cross of two intermediates gives 1:2:1.	Using a 3:1 ratio, or mixing up blended and both-shown.	A heterozygote that differs from both parents.
6	X-linked inheritance	Sons get their only X from their mother; a carrier mother and an unaffected father have a 1/2 chance per son of an affected son and no affected daughters.	Reporting the risk for all children, or thinking an unaffected father protects sons.	A carrier mother and a trait like hemophilia or color blindness.
7	Hardy-Weinberg frequencies	$p + q = 1$ and $p^2 + 2pq + q^2 = 1$; the recessive phenotype frequency is q^2 , so $q = \sqrt{\text{that frequency}}$.	Using the phenotype percentage as q or forgetting the 2 in $2pq$.	A percent showing the recessive trait and “how many carriers?”
8	Homologous vs. analogous structures	Homologous structures share an underlying plan from common ancestry, even if used differently; analogous structures share a function but evolved separately.	Classifying by function rather than by origin.	Same bones, different uses; or same use, different build.

Course scope (public description): This course is a foundational introduction to the biological sciences. This course explores the overarching theories of life from biological research as well as the fundamental concepts and principles of the study of living organisms and their interaction with the environment. Key concepts include how living organisms use and produce energy; how life grows, develops, and reproduces; how life responds to the environment to maintain internal stability; and how life evolves and adapts to the environment.